

Python Complete Notes (Beginner to Intermediate)

1. Variables

A variable is a container used to store data values.

```
x = 10  
name = 'Vamsi'
```

2. Data Types

Defines the type of value stored.

```
a=10  
b=10.5  
c='Hello'  
d=True  
l=[1,2]  
t=(1,2)  
s={1,2}  
d={'name': 'Vamsi'}
```

3. Operators

Used to perform operations.

```
a+b  
a>b  
a and b  
x+=5  
2 in [1,2,3]  
a is b
```

4. Conditional Statements

```
if marks>=50:  
    print('Pass')  
else:  
    print('Fail')
```

5. Loops

```
for i in [1,2,3]: print(i)  
  
i=1  
while i<=5:  
    print(i)  
    i+=1
```

6. Functions

```
def add(a,b):  
    return a+b
```

7. Scope

```
x=100

def fun():
    print(x)
```

8. Inbuilt Functions

```
len([1,2])
sum([1,2])
max([1,2])
```

9. OOPS

```
class Student:
    def __init__(self,name):
        self.name=name
```

10. 4 Pillars

Encapsulation, Inheritance, Polymorphism, Abstraction

11. Modules

```
import math
math.sqrt(16)
```

12. Errors

SyntaxError, NameError, TypeError, ValueError, IndexError, KeyError, ZeroDivisionError, AttributeError

13. Error Handling

```
try:
    x=int('abc')
except:
    print('Error')
```

14. File Handling

```
with open('file.txt','r') as f:
    print(f.read())
```

15. CRUD Operations

```
l=[1,2]
l.append(3)

d={'a':1}
d['b']=2
```

16. Important Methods

```
s.upper()  
l.append()  
d.keys()
```